

Orchestrating Microbiome Analysis with Bioconductor

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Abstract

The expansion of microbiome research has led to the accumulation of interlinked datasets encompassing versatile taxonomic and functional assays. While critical to advance the field, the analysis of increasingly large and heterogeneous multi-modal microbiome data would benefit from unified approaches supporting interoperability between methods and the design of modular data science workflows. The Bioconductor project has recently developed an optimized statistical programming framework for multi-assay data integration in computational life sciences. Building on this foundation, we introduce a community-developed open-source ecosystem for microbiome data science, dedicated to supporting joint analysis of hierarchical, interlinked, and heterogeneous datasets that are increasingly common in modern microbiome research. This interoperable infrastructure comprises of open-source datasets, methods, tutorials and an active online community, allowing analyst to perform data wrangling, analysis and visualization in a coherent and standardized manner. Its functionality and usage are detailed in the OMA book <https://microbiome.github.io/OMA>, which offers guidance for prospective users and contributors.

Keywords: microbiome, Bioconductor, data science, data integration

Introduction

Modern data science applications critically rely on open-source methods created by the research community^{1;2} and open software ecosystems have become central innovation hubs, driving technological and cultural transformation³. The Bioconductor project has been established as a major distribution channel for open data science methods in the life sciences. Nowadays, its vast developer community maintains over 2,300 quality-controlled R packages for various fields of life science informatics⁴. Additionally, Bioconductor has evolved into a global community that supports data science collaboration and training⁵, aligning with recent international calls to enhance microbiology education in particular⁶.

Microbiome research focuses on how microbial communities vary, function and interact with their environment and potential host⁷. The technologies used in this field generate vast amounts of high-dimensional and hierarchically structured data, commonly in form of microbial DNA sequences detected in a list of samples. The rapid expansion of microbiome research has led to the accumulation of interlinked, multi-modal datasets encompassing diverse taxonomic and functional assays, ranging from microbial abundances and phylogenetic relationships to functional profiles. Microbiome data scientists are thus increasingly facing the practical challenges of integrating heterogeneous data across multiple data modes into reproducible workflows. The unique statistical properties and complexity of microbiome data necessitate specialized approaches in their analysis⁷⁻⁹. While the diversity of proposed solutions can make it difficult to identify the optimal methods and build interoperable workflows^{10;11}, for

example, different approaches for indicator taxa analysis often produce inconsistent results^{12–14}. Thus, the need for standardized microbiome data science infrastructures has been widely recognized⁷ and a variety of open microbiome data science frameworks have been developed, including Anvi'o¹⁵, Mothur¹⁶, QIIME2¹⁷, and phyloseq¹⁸. However, platforms focused solely on microbiome data science struggle to address the need to integrate data across diverse modalities and research domains, from metagenomics to metabolomics.

The Bioconductor project embeds the methods of microbiome data science into a wider statistical programming framework alongside the concurrent development of interoperable methods and workflows for transcriptomics¹⁹, single-cell²⁰, metabolomics²¹, proteomics^{21;22}, and other fields. Here, we introduce a community-developed open-source framework, that supports the integration of multi-modal microbiome datasets. It provides harmonized data presentation, streamlined data import, and fast, robust methods for transforming raw abundance tables into biological insights. The methods are extensively tested and distributed through R/Bioconductor packages, references being available in <https://bioconductor.org/packages>. To help users to leverage the framework and promote evidence-based best practices, we present online book <https://microbiome.github.io/OMA> that features installation, sources of help and dedicated topics for specific aspects of microbiome analysis.

Data infrastructure

Data scientists routinely deal with interlinked assays, hierarchical data (*e.g.*, ontologies, trees), relational data (*e.g.*, networks), and additional side information. This side information can include sample attributes like treatment group, collection site, or time point. The ability to integrate heterogeneous data elements is essential in biological research, where the individual components of a system can be seldom studied in isolation. However, input data formats vary across different methods, leading to technical complications and time lost on data wrangling. Bioconductor addresses this through standardized data structures supported by an ecosystem of interoperable methods, allowing more time to be invested into core analytical tasks.

Bioconductor data structures support statistical analysis of complex data combinations²³ and are widely adopted by various research communities. They have been implemented through specialized object-oriented classes that integrate multiple data elements into a single structure known as a *data container*²⁴. The Summarized-Experiment (SE) class is the primary data container underlying the Bioconductor ecosystem^{23;25}. Its position among the top-1% most downloaded packages in Bioconductor reflects its widespread recognition and increasing adoption among developers (Figure 1). SE provides a standardized solution to store tabular data by linking numeric abundance matrices with side information on features (rows) and samples (columns). Side information may include sample metadata (*e.g.*, origin, collection time, host health) and feature metadata (*e.g.*, taxonomy).

The interoperability of SE with other Bioconductor data structures facilitates the transfer of methods across different fields of omics research, yielding improved overall quality control and user support. This is a major strength, which has facilitated the development of modular and scalable data science methods⁵. Consequently, the SE-based framework has been extended to similar data integration tasks, notably in single-cell and spatial omics through the SingleCellExperiment (SCE)²⁰ and SpatialExperiment¹⁹ classes.

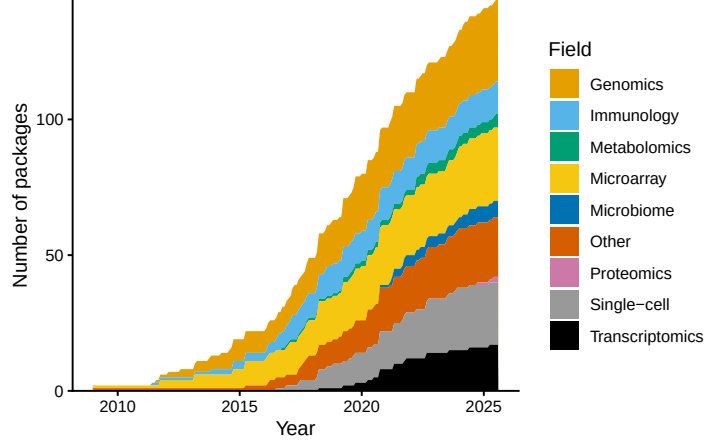


Figure 1: Adoption of the SummarizedExperiment (SE) data container among Bioconductor developers. The growth in the number of Bioconductor packages supporting SE over time is shown, categorized by their respective fields. For some of the packages SE support has been added after their original release.

Hierarchical data structures. Microbiome data scientists frequently need to integrate information on feature phylogenies and sample hierarchies in order to accurately reflect fine-scale variability in microbiome data. While maintaining interoperability with the broader Bioconductor ecosystem, the TreeSummarizedExperiment (TreeSE) class²⁶ extends the SE and SCE data structures to hierarchical datasets by incorporating both feature and sample organisations as tree structures (Figure 2 a). Moreover, TreeSE supports the storage of other common aspects of microbiome data, for example, reference sequences, and incorporates results from common analytical procedures such as dimensionality reduction, statistical transformations, and agglomeration by taxonomic, functional or other information. Discussed later, the MultiAssayExperiment (MAE) class offers even more flexibility for data integration (Figure 2 b). The interoperability with other Bioconductor classes facilitates scalable analysis and the design of modular yet flexible workflows. The direct interoperability of TreeSE with the widely adopted SE data science ecosystem distinguishes our TreeSE-based approach from phyloseq, which is another popular Bioconductor data container originally developed with a focus on 16S amplicon data analysis¹⁸. TreeSE is a superset that can accommodate all elements of a phyloseq object (taxa abundances, sample metadata, and

phylogenetic tree) while also accommodating multi-modal datasets, including functional profiles, metabolomics measurements, or alternative feature representations. Additionally, it offers often substantial gains in memory efficiency and computing speed (Figure 3). Overall, the TreeSE approach outperforms phyloseq in benchmarks. While the phyloseq extension speedyseq is faster in certain operations, TreeSE scales more efficiently in both execution time (t) and allocated memory (m) as sample size (n) increases. Furthermore, conversion functions and detailed cheat sheets between the two formats are available to bridge the gap between the respective user communities.

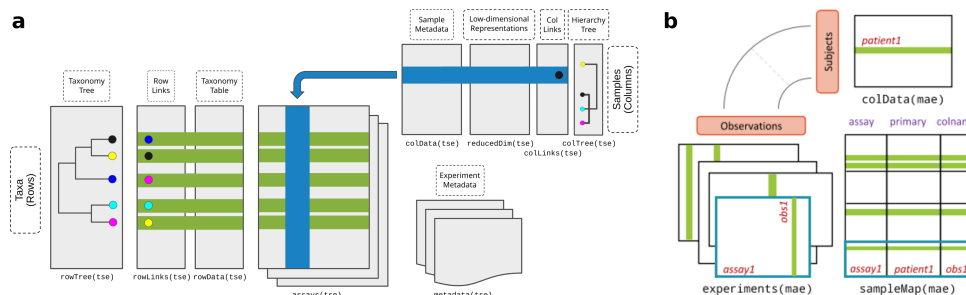


Figure 2: Bioconductor data containers for microbiome data. (a) **TreeSummarizedExperiment (TreeSE)**. TreeSE supports hierarchical and multi-table analysis in microbiome data science²⁶. It links the numeric abundance matrix and potential alternative versions (statistical transformations, different taxonomic levels, dimensionality reduction) with tabular and hierarchical side information on the features (rows) and samples (columns). Additional slots for reference sequences and experiment metadata are available. (Figure modified from²⁶) (b) **MultiAssayExperiment (MAE)**. MAE facilitates integration of data from multiple experiments²⁷. (Figure from²⁷)

Multi-assay data structures. When integrating data across multiple modalities, such as taxonomy, metabolomics, or host genomics, more flexible sample mapping can be very useful. Multi-omics data is often sparse, and samples can be only partially matched between experiments. In other cases, a single sample in one modality can be matched by two or more samples in another modality see²⁹. The MultiAssayExperiment (MAE) data container²⁷ facilitates the integration of multiple data objects representing different omics modalities (see Figure 2 b). Whereas TreeSE is restricted to a one-to-one sample mapping between different data tables, MAE supports flexible mappings between multiple experiments from the same study. Integration of multi-table data without such a structured framework often leads to ambiguous mappings and inconsistent results. For instance, diagonal integration methods^{30;31} — where shared features across data types may not be explicitly defined — can benefit from such a systematic framework that enforces anchor alignment. This mirrors lessons learned from single-cell data integration, where shared anchors and hierarchical mapping have proven essential for interpretability. Flexible mapping of samples across

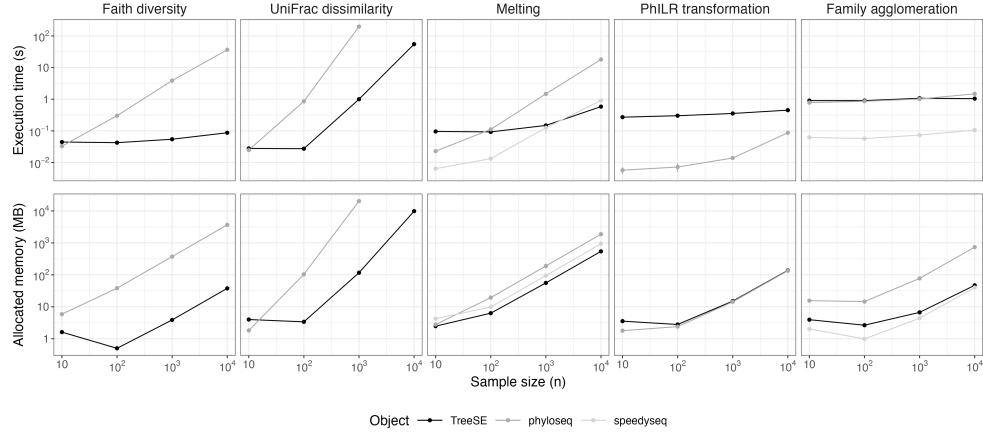


Figure 3: Comparison of execution time and memory consumption between alternative methods in Bioconductor TreeSE, phyloseq and speedyseq were benchmarked for five common analytical operations applied to random subsets of samples from a large study on wild baboons²⁸ (accessible through the microbiome-DataSets Bioconductor package). Execution time (s) values and error bars represent the mean and standard error (SE) over 10 iterations, respectively. Allocated memory (MB) was estimated based on a single iteration. Overall, TreeSE outperformed phyloseq. While speedyseq achieved higher performance in the supported operations, TreeSE scaled more efficiently in both execution time and memory allocation with increasing sample size (n). The benchmarking analysis was conducted in R using the bench library with 8 CPU cores and 32 GB RAM. All operations, except for UniFrac dissimilarity with phyloseq, could also be performed on a regular laptop (*e.g.*, 4 CPU cores and 16 GB RAM). The source code for the benchmarking experiments can be accessed through the OMA online book.

several data objects supports efficient data integration, methods sharing, and overall interoperability.

Table 1 summarizes the microbiome-related data containers, while Table 2 introduces the available data elements in each container.

Table 1: Recommended data containers for microbiome analysis.

Data container	Target use
SummarizedExperiment (SE) ²³	generic data container for experiments with multiple assays
TreeSummarizedExperiment (TreeSE) ²⁶	tailored data container for hierarchically organized experiments
MultiAssayExperiment (MAE) ²⁷	tailored data container for sets of multi-omic experiments

Table 2: Key data elements in the employed data containers.

Slot	Content	SE	TreeSE	MAE ^a
Tables				
assays	tables of features (rows) and samples (cols)	X	X	X
altExps ^b	experiments with variable number of features		X	X
reducedDims	results of dimensionality reduction		X	X
Rows				
rowData	side information on features	X	X	X
rowPairs	linkage between pairs of associated features		X	X
rowTrees	hierarchical organization of features		X	X
referenceSeq	reference sequences of features		X	X
Columns				
colData	side information on samples	X	X	X
colTrees	hierarchical organization of samples		X	X
sampleMap ^c	flexible sample mapping across experiments			X
Other				
metadata	additional information on the experiment	X	X	X

^a While MAE does not contain conventional slots (with the exception of colData), its experiments can include any number of data containers that provide those slots.

^b altExp samples must match assay samples in a 1-to-1 mapping. ^c sampleMap supports diverse mappings between samples from different experiments (1-to-1, 1-to-many, many-to-1 and many-to-many).

Data resources. Several open microbiome data resources are supported, enabling direct data import into the aforementioned data containers for further downstream analysis within the Bioconductor methods ecosystem (see Table 3). Examples of the available data resources include curatedMetagenomicData³², HoloFood³³, microbiomeDataSets³⁴, and the EBI/MGnify database³⁵. In addition, Bioconductor packages often include built-in demonstration datasets. Collectively, these resources encompass taxonomic and functional profiles from thousands of published microbiome studies and tens of thousands samples that can be accessed with the available tools for research and educational purposes.

Table 3: Open microbiome data resources supporting the framework (accessed 27/09/2025).

Data resource	Package	# studies or datasets	# samples
Curated metagenomic data ³²	curatedMetagenomicData ³²	93	22,588
HoloFood ³³	HoloFoodR ³⁶	-	9,990
MGnify ³⁵	MGnifyR ³⁷	5,129	629,821

Data resource	Package	# studies	
		or datasets	# samples
microbiomeDataSets ³⁴	microbiomeDataSets ³⁴	6	19,100

Data preparation

While microbiome data scientists continue to develop specialized solutions for data analysis^{8;9;38}, the implementations typically differ in their expected input and output formats. Common data standards support modular data science workflows, speeding up methods development, benchmarking, and replication. Here, we describe a recently emerged ecosystem of software packages that support microbiome data integration and analysis based on the data containers, (Tree)SE and MAE (see Figure 4). On top of this package ecosystem we introduce a framework for microbiome data analysis.

The framework is focused on the downstream analysis of high-throughput sequencing data from microbiome experiments after the original measurements have been summarized into abundance tables (*e.g.*, taxonomic or functional profiles). Taxonomic abundance tables are one of the most commonly encountered data types. Common measurement platforms include amplicon-based marker gene studies (*e.g.*, 16S rRNA) and metagenomic sequencing, but alternative profiling techniques are available including phylogenetic microarrays³⁹ and high-density optical mapping⁴⁰. The interoperability of the data containers support modular workflow design for integrating different omics data into the wider Bioconductor ecosystem. This decisively mitigates reproducibility challenges posed by rapidly evolving reference databases and profiling tools (*e.g.*, MetaPhlAn2 vs. MetaPhlAn4) by enabling structured coexistence and direct comparison within a unified, version-resilient framework^{41;42}.

Data import. The TreeSE data container can be populated with microbiome data from common tabular data types but non-standard file formats may require additional data wrangling. To streamline data import, we provide importers for a variety of standard microbiome file formats including BIOM⁴³, MetaPhlAn/HUMAnN (MetaPhlAn v. 2-4, HUMAnN v. 3)⁴⁴, Mothur¹⁶, QIIME2¹⁷ and the TAXonomic Profile Aggregation and STandardisation format (taxpasta)⁴⁵. These importers bridge the gap between the raw sequence processing and statistical analysis in R, allowing researchers to focus on interpreting biological patterns rather than wrangling file formats. Moreover, converters between alternative data structures in R are available supporting seamless conversions between TreeSE and BIOM, DADA2⁴⁶ and phyloseq (the `mia::convert*` functions), and TreeSE-based microbiome data objects can be exported into external services such as MicrobiomeDB⁴⁷ and in other languages (*e.g.*, Julia). These converters facilitate widespread access to microbiome analysis methods.

Quality control and exploration. After importing the data into R, the recommended first step involves basic data exploration and quality control⁴⁸. In addition to methods for identifying contaminant sequences and calculating summary statistics, the framework includes a versatile set of custom visualization methods to assess data variability, outliers, homogeneity of variance and other aspects that may need to be

considered in downstream data analysis. The *mia* and *miaViz* packages provide convenient access to many operations for common exploratory tasks in microbiome analysis and visualization for *TreeSE* objects (Figure 5).

Filtering and agglomeration. Microbiome datasets can contain tens of thousands of unique features depending on the sequencing resolution, used reference databases, and the scope of analysis. To summarize data into biologically meaningful groups or to reduce multiple comparisons the data is often subsetting, filtered or agglomerated to higher-level taxonomic or functional categories or stratified by sample groups. The package ecosystem and the *mia* package in particular feature various tools for such tasks. For instance, the prevalence based methods can be useful to remove rare features that are likely sequencing artifacts or present otherwise limited value for the analysis. The mechanism of alternative experiments (*altExp*) supports data agglomeration by taxonomic levels and by other features such as pathogenicity or functional potential (*e.g.*, butyrate production⁴⁹, gut-brain modules⁵⁰, and other host-associated signatures⁵¹) and storing them as parallel data tables. The supporting methods incorporate the agglomerated levels within the same data object for parallel operations, and can take care of agglomerating the associated tree information and alternative assays. Thus, filtering, subsetting, and agglomeration operations are supported in both feature as well as sample space, and simultaneously propagate to cover the relevant alternative assays, alternative experiments, ordinations, and side information. This supports the efficient management of interlinked assays and reduces repetitive operations on data.

Transformation. Statistical transformations commonly used in microbiome research include relative abundance (*e.g.* TSS)⁵², tree-aware phylogenetic ILR (*phILR*)⁵³, centered log-ratio (CLR), and robust centered log-ratio (*rCLR*). The latter two of which aim to mitigate compositionality bias⁵⁴. Data rarification is a related concept that has been suggested to control for uneven sequencing depths, however it remains widely debated in microbial ecology^{55;56}. Our framework supports these and a variety of other transformations commonly applied to microbiome data.

Data enrichment. Incorporating additional data, for example, new versions of feature abundance data resulting from data transformations, is a frequent operation in applied data analysis, and it often takes place in an iterative fashion. The data containers support gradual and organized accumulation of interlinked data elements including transformations, aggregated data, dimension reduction, and derived data on samples and features. We generally recommend that the data objects are initialized at the beginning of the workflow, where they can then be gradually enriched as the project advances. By systematically extending the upstream data preparation step — instead of ad-hoc calculations scattered amidst the workflow — one can ensure well-organized and transparent records of data transformations. Data can be enriched by importing new external information as well as by deriving new information from the data itself into the same data container as described in the next section.

Downstream data analysis

This new Bioconductor framework for microbiome data science improves the management of integrative analysis workflows. It facilitates systematic linking of the different data elements and analysis outcomes, helping to shift the focus towards multi-modal analyses and methods development while supporting standard analysis tasks in microbiome research.

Microbiome data analysis often relies on specialized statistical approaches as data sparsity, zero-inflation, compositionality, and heteroschedasticity pose challenges for standard data analytical methods^{8;52}. Moreover, microbiome data is hierarchically structured across taxonomic, ecological, temporal, and spatial levels^{7;57}. Alternative processing pipelines introduce different biases and generate varying data formats; for example, the same sample can yield differing taxonomic profiles depending on the computational method used⁵⁸. These and other unique characteristics have wide-reaching implications for microbiome analysis, necessitating dedicated analysis methods⁷. Our statistical programming framework supports the community-driven development and benchmarking of the solutions. The following sections provide an overview of the available tools.

Community indicators. Various general indicators are available to quantify the microbial community state or other sample properties. Community diversity represents a common measure in microbial ecology⁵⁹ (Figure 5). Alpha diversity metrics generally quantify the observed or estimated number of distinct taxa (“richness”) and their distribution (“evenness”). Many diversity indices (*e.g.*, Shannon index), combine these two aspects. Phylogenetic diversity measures additionally consider the breadth of phylogenetic relatedness of community members⁶⁰; the new implementation of the Stacked Faith’s Phylogenetic Diversity can accelerate the computational performance by 1-2 orders of magnitude⁶¹. Enhanced analyses with repeated rarefaction are supported⁶² albeit the use of rarefaction remains controversial^{56;63}. Other global indices of community state include measures of dominance, rarity, skewness, kurtosis, modality, library size (number of reads), or absolute abundance quantification. Sample information in the TreeSE object (colData) can be enriched with these derived quantities. By default, the `mia::addAlpha` method returns a set of complementary alpha diversity measures⁶⁴.

Community similarity is commonly evaluated with beta diversity measures, which quantify multivariate dissimilarity between feature communities across all pairs of samples and can be used to quantitatively compare community composition. The framework supports common (dis-)similarity metrics in microbial ecology including Bray-Curtis, Jaccard, Aitchison^{54;59}, and the phylogeny-aware UniFrac⁶⁵. The community dissimilarities are commonly visualized with unsupervised ordination methods that project the data onto a lower-dimensional display⁶⁶. Unsupervised techniques, Principal Component Analysis (PCA) and Principal Coordinates Analysis (PCoA) also known as Multi-Dimensional Scaling (MDS), and Non-metric MDS (NMDS) (Figure 5) are commonly used. Statistical tests (*e.g.*, Permutational Multivariate Analysis of Variance) and supervised visualization methods, such as the distance-based

redundancy analysis (dbRDA) can complement the unsupervised analyses, and help to uncover and quantify co-variation between community composition and sample-specific covariates. This can benefit from compositionally robust measures (*e.g.*, robust Aitchison distance⁵⁴), or phylogeny-aware dissimilarity measures specifically developed for microbiome research (*e.g.*, Unifrac⁶⁵). Sample dissimilarities provide the basis for many specialized microbiome techniques including microbial community typing (clustering) with Dirichlet Multinomial Mixtures (DMM)⁶⁷ and community state types (CST)⁶⁸, or factorization into community signatures⁶⁹. These and other multivariate methods are provided by *mia*, *miaViz*, and additional single-cell packages (*scater*, *scuttle*²⁰).

Differential abundance and mediation analysis. Differential abundance analysis (DAA) and differential prevalence analysis (DPA) methods are used to identify indicator microbial features whose abundances vary across study groups or continuous gradients, for example, spatial or temporal distance between samples (Figure 5). Many dedicated DAA methods in microbiome research support the TreeSE framework including methods such as ANCOM-BC2⁷⁰, LimROTS²², MaAsLin3⁷¹, and radEmu⁷². Recent studies by us and others indicated that elementary methods, such as log transformed total sum scaled (TSS) counts coupled with a linear model, or presence/absence data with logistic regression often yield the most consistent results¹³. The standard DAA typically focuses on individual features, ignoring interactions or other covariation among feature abundances. Thus, we support covariation analyses; indeed, agglomerating collinear features as a preprocessing step can reduce multiple testing, increase statistical power, and support interpretation but is also prone to subjective parameterization and limited generalizability. The grouping can be based on clustering of the abundance profiles or on external information drawn from phylogenetic relations, taxonomic classifications or functional properties (*e.g.*, lump together all butyrate producers⁴⁹). Beyond these, advanced methods — such as causal mediation analysis tailored to microbiome multi-omics data — are being developed using SE⁷³, highlighting its readiness to support emerging innovations in microbiome analytics.

Function- and sequence-level analyses. Taxonomic analysis is frequently complemented by functional analyses, either based on functional predictions derived from amplicon or metagenomic sequences, or from parallel metabolomic or other functional assays. Our framework not only supports incorporation of functional abundance tables within the same data objects but specialized methods on functional data are available; for instance, regarding metabolome analyses the *notame* package⁷⁴ recently added SE support, and the R4MS project provides many interoperable tools²¹. The framework has found recent applications in extended analyses of metagenomic features, for instance, antimicrobial resistance gene composition, diversity, and load⁷⁵. Following this, we have converted a large-scale compilation of resistome profiles in 8,972 metagenome samples⁷⁶ in the TreeSE format for research and educational purposes.

Spatio-temporal, ecological and epidemiological models. Microbiome time series⁷⁷ encompass a wide range of study designs, ranging from short to long follow-up times, sparse to dense data, univariate to multivariate and multi-omic monitoring⁷⁸,

and real, pseudo, or simulated time series. Moreover, an increasing number of studies have incorporated the spatial dimension in microbiome analysis across scales⁵⁷. While spatial and temporal analyses require specialized methods, the miaTime R package offers dedicated methods for longitudinal microbiome studies. In addition, Bioconductor contains applicable methods from other research domains, namely single-cell sequencing²⁰ and spatial transcriptomics¹⁹, which use closely related data containers (Figure 5). Spatial and temporal information can be incorporated as a covariate (*e.g.*, location or time) or by estimating temporally or spatially resolved metrics, such as divergence (dissimilarity between consecutive time points), auto-correlation (similarity of closely related samples), and feature modality (taxa with distinct low and high abundance states). Another type of temporal analysis includes the time-to-event, or survival models, which are encountered regularly in, epidemiological and population cohort studies; for microbiome data, these analyses require specialized approaches, as proposed in previous studies^{79;80}. We also provide methods to simulate time series from standard models of microbial community dynamics including Hubbell’s neutral model, generalized Lotka-Volterra, and the consumer-resource model through the miaSim package⁸¹.

Multi-modal integration. An increasing number of studies integrate microbial abundance data with further omics, including transcriptomics, proteomics, and host genomics^{7;82–84} (Figure 5). As demonstrated above, Bioconductor data containers support combinations of complex omics data^{20;23;27}. Available computational approaches for multi-table analysis in microbiome research can be broadly categorized into three groups: Machine Learning (ML) predictive modelling⁸⁵, association-based modelling, and latent variable modelling^{7;8;85–88}. Some of the currently supported multi-omic methods include cross-correlation analyses, canonical correspondence analysis (CCA), anansi⁸⁷, and joint-RPCA [in press].

Other methods and programming environments. A range of other methods is employed in microbiome data science. Examples include network analysis⁸⁹, ML-based prediction and classification⁹⁰, enrichment analyses via BugSigDB⁵¹ or Microbe Set Enrichment Analysis (MSEA)⁹¹. The recent adoption of interoperable data containers and the methodological similarities in related fields can greatly expand the selection of available toolkits in emerging areas. In addition to the internal coherence within the Bioconductor ecosystem, this framework supports integration with external platforms through interfaces, namely `reticulate` for Python, `Rcpp` for C++, and compatibility with pipelines bioBakery^{44;92}. These connections empower users to combine the best tools available across environments and languages while maintaining structured data interoperability via standard containers like TreeSE and MAE.

Accessible analysis and community

Modern science is embodied by a critical community capable of assessing discoveries and replicating results⁹³. Open software and the social structures around it present a timely manifestation of this principle, with technical, legal, sociological and epistemic consequences³. FAIR principles for software support access to well-documented

and user-friendly methods¹, and are facilitated by the extensive quality control and documentation standards enforced by Bioconductor²³. In addition to these technical demands, the development and quality control of academic software critically relies on active critical source communities that can provide continuous benchmarking and support the identification and dissemination of evidence-based best practices^{1;3;77}. We actively support community formation by online training resources and communication channels. Together with the interactive applications, these resources provide accessible entry points for prospective developers and users.

The OMA book. *Orchestrating Microbiome Analysis with Bioconductor (OMA)* (<https://microbiome.github.io/OMA/docs/devel/>) is an online book that provides a comprehensive introduction to the microbiome data science framework in Bioconductor. It offers practical guidance, reproducible workflows, and training material for integrative microbiome analyses. The book is versioned to match Bioconductor’s biannual software release cycles. The work incorporates rich feedback from the user community and microbiome training courses. It complements the broader trend, where specific Bioconductor data structures^{23;26;27} have been adopted as a solution for multi-table data integration tasks in various domains such as single-cell sequencing (OSCA book)²⁰ and transcriptomics [OSTA book](#). These resources demonstrate how standard data containers can support statistical programming on interlinked multi-table data sets in a highly optimized way.

Interactive applications. Graphical user interfaces provide accessible entry points to the framework. The iSEEtree and miaDash packages provide interactive analysis and visualization tools for microbiome data⁹⁴ (Figure 5). Their interface supports automatic generation of source code and replicable workflows as well as analysis templates for users without a strong programming expertise. Moreover, the tidy R paradigm, available through tidyomics, simplifies the syntax and can streamline the workflows⁹⁵. These tools allow users to perform basic analyses and visualize data with ease.

Community. Its widespread adoption has made R one of the most cited software resources of our time² and the introduction of the shared data structures facilitates the broad interoperability of the methods ecosystem. Bioconductor has a strong tradition in fostering the R user community through shared data structures and training activities⁵. An active community of developers and users thrives online, supporting the software sustainability and promoting good practices in scientific computing⁹⁶. The community interacts via several online communication channels, including Bioconductor Zulip, GitHub platform, and an email list. The development of the framework is influenced by feedback from the user community through these channels, regular meetings and training workshops.

Outlook

The Bioconductor community has a well-established reputation in developing open, peer-reviewed software for life sciences. Its strengths include the enforcement

of minimal documentation standards, semantic versioning, portability across operating platforms, quality control (*e.g.*, unit testing, continuous integration), usability (APIs), and robustness (deprecation mechanisms), and a biannual release cycle. During the past two decades Bioconductor has grown into a repository of 2,300 actively maintained software packages, of which 60 are classified under the Microbiome Bioc task view. In order to enjoy the benefits of automation while maintaining flexibility, data scientists routinely remix open-source software into custom workflows. However, the lack of commonly adopted data representation standards often hinders the development, sharing, and application of statistical workflows. At the core of the Bioconductor’s data science strategy is the idea of shared data representation. The enhanced interoperability can help to minimize redundant efforts and facilitate the translation of methods between different research areas^{1;97} and, importantly, facilitates the formation of broad developer and user communities.

Microbiome research has shifted towards integrative analyses aiming at mechanistic and causal inference^{59;98}. Technological advances in long-read sequencing, functional assays, spatial profiling, and single-cell analysis have created an increasing demand for integrative data analysis techniques. Thus, microbiome data is increasingly linked with other omics along varying spatial and temporal dimensions. Our framework addresses this need, together with parallel developments in other areas of omics.

Recent developments in microbiome and multi-omics workflows increasingly rely on Bayesian modeling and Gaussian processes to accommodate uncertainty and temporal or spatial structures⁹⁹. Our framework is compatible with these approaches, providing the structured input necessary for probabilistic inference and scalable computation. Building on these statistical foundations, the rapid emergence of foundation models and deep learning approaches in biomedicine calls for robust, scalable data representations¹⁰⁰. Our framework, through its use of standardized containers and modular workflows, is well-positioned to support such advances. Applications include transfer learning across studies, interpretable embeddings of microbiome multi-omic profiles, computational prediction of microbial function and metabolites^{101;102}, and integration with AI agents for hypothesis generation and automated analysis, among others^{103;104}.

Open data science infrastructures have become critical components of reproducible research and a vast body of applications critically rely on open-source methods and workflows^{1;3;77}. Standardized data structures can enhance the transparency, reliability, and explainability of analysis workflows. We have documented the interoperable framework built on widely adopted data containers, open data resources, analysis methods and reproducible workflows for multi-modal microbiome research. Further developments could include extended methodology for temporal, spatial, and multi-omic analyses. Supporting interoperability between R and other programming languages can facilitate the application of other ML frameworks. As the field evolves, Bioconductor will continue to integrate new tools and approaches to improve integrative analyses of microbiome data.

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Contributions

LL and TB initiated and coordinated the work. All authors contributed material, participated in manuscript preparation, and approved the final version.

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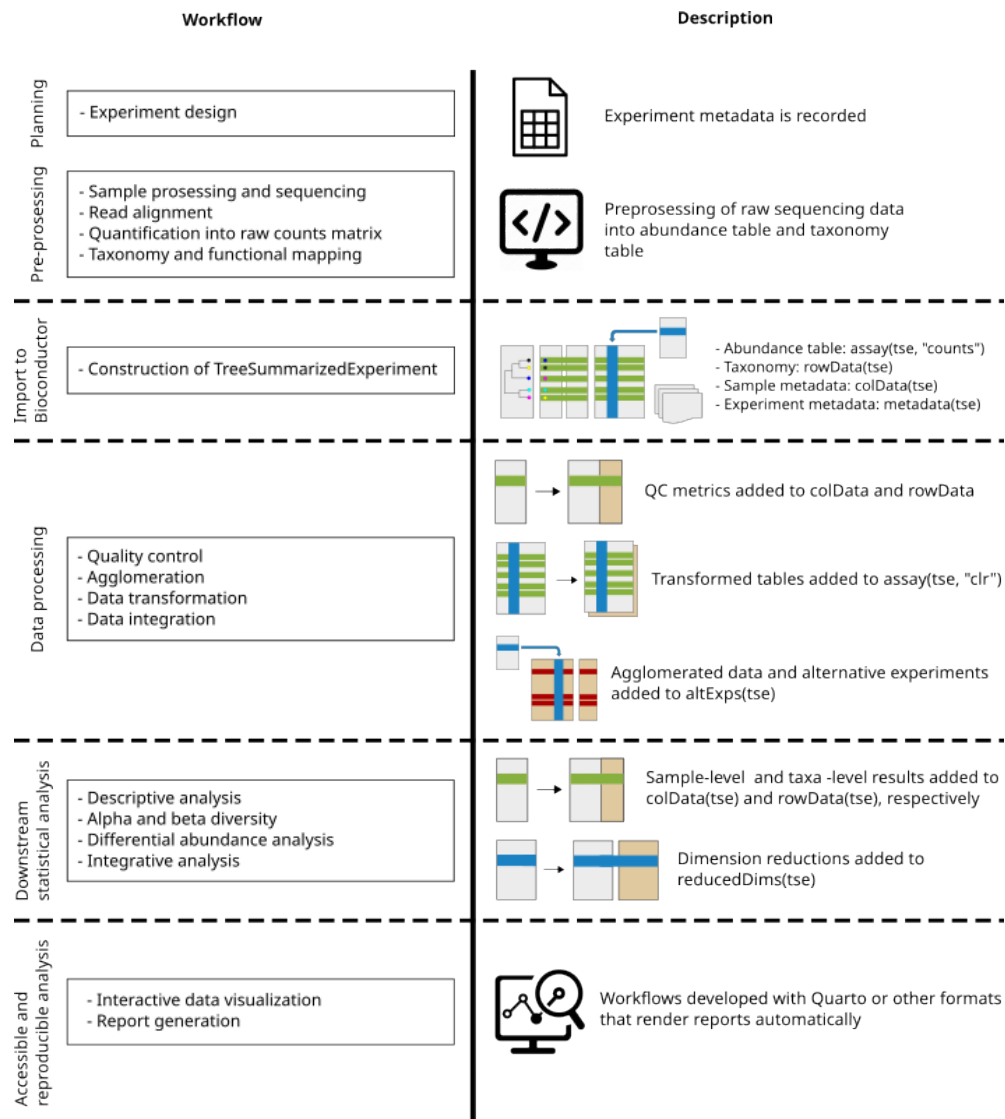


Figure 4: Examples of Bioconductor tools along the microbiome data science workflow. A typical microbiome workflow starts with taxonomic and functional annotation. The processed microbiome data can then be imported to Bioconductor data container (TreeSE), or integrated with other omics using the MAE container for downstream statistical analysis. (Figure adapted from ²⁰)

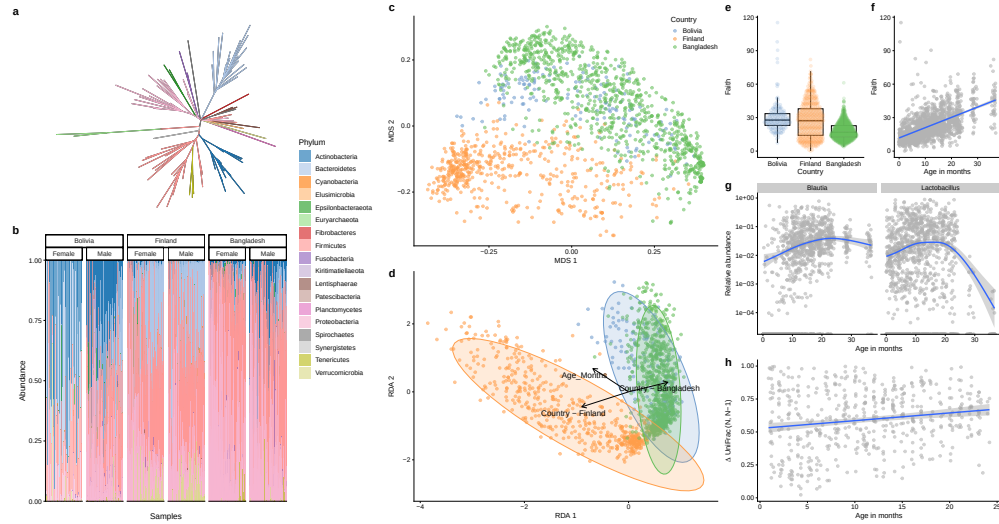


Figure 5: Common visualization methods applied to time series data on infants from three different countries. **a**, Phylogenetic relationships. **b**, Phyla abundances of the samples ordered by age, stratified by country and gender. Actinobacteria is more present in Bolivia and Bangladesh compared to Finland. **c-d**, Multidimensional Scaling (MDS) using Unifrac distances and Distance-based Redundancy Analysis (dbRDA) based on Aitchison distance, respectively. Both ordination methods shows clear pattern separating microbial profiles in different countries. dbRDA shows additionally association between microbial profile and age. **e-f**, Faith's phylogenetic diversity by country and age respectively. **g**, Differential Abundance Analysis (DAA) results. Two genera exhibit association with age. **h**, Unifrac divergence between consecutive time points. Results indicate slow stabilization of microbial communities.